SILC Four-Model Forest Growth Benchmark

Companion Report v10 — 4-model year 100 + refined inputs

Aaron R. Weiskittel, CRSF, University of Maine — May 2026

Executive summary

Three findings drive this update.

- (1) The +18.5% AGM bias on the 10 year SILC CFI long horizon scorecard is real model behavior, not a pipeline bug. Triple checking the AGM treelist to BA / Cords / BdFt computation reproduces to 1e-13 ft^2/ac, and observed STAND_METRICS BA matches direct tree list recompute to 0.005 ft^2/ac. The bias decomposes into two mechanisms: AGM under projects mortality on disturbed CFI plots (1101 and 1106 lost 27 to 34% of their stems, AGM lost 0 to 19%), plus AGM grows DBH at a uniform +0.8 in per decade regardless of site (plots 1100 and 1104 grew +0.05 to 0.17 in observed).
- (2) Switching on the #126b in-source MORTCAL correction in AcadianGY 12.3.9 (default OFF) cuts the 10 year BA bias from +18.5% to +7.8%, RMSE -34%, R^2 from -0.20 to +0.47. Sawlog BdFt bias drops to +4.0% with R^2 = 0.84, the best of any model on this metric.
- (3) Pushed to 100 years, MORTCAL pulls projected BA from a ceiling around 200 ft^2/ac down to a credible carrying capacity of 98 to 108 ft^2/ac. Year 100 NetCords lands at 35 to 39 cords/ac across all six byStrata cells, plus CFI plot backfill at Cedar A+B (35.8 cords/ac) and Mixedwood A+B (21.9 cords/ac).
- (4) Full 4 model year 100 view added: FVS-NE default 209 to 291 BA, FVS-NE calibrated 125 to 225, OSM-ACD 172 to 221 (covered cells). AGM MORTCAL still the lowest projection. Refined inputs from CFI v3 lat/long swap BGI 3000 to 3902 and CSI 12 to 15.78 m; minimal effect on relative ranking.

Recommended operational stack: AGM with MORTCAL on as the long horizon operational anchor (35 to 39 cords/ac year 100); FVS-NE calibrated as the upper bracket (44 to 70 cords/ac); OSM-ACD as additional cross check where coverage exists.

Triple check of AGM long horizon calculations

The +18.5% AGM bias at the 10 year CFI scorecard prompted a forensic audit. Every step of the pipeline checked out.

Source code: ~/AcadianGY_12.3.9.r on Cardinal (identical to local v12.3.5 plus the opt-in MORTCAL correction). AcadianGYOneStand is annual (cyclen=1 hardcoded since 2022-09-01). The driver loops PERIOD_YR times, giving the correct horizon.

Pipeline verification checks

Check	Result
AGM treelist to BA/Cords/BdFt reproduces results CSV	exact to 1e-13
STAND_METRICS BA_CURR vs direct TREE.csv recompute	match to 0.005 ft^2/ac
AcadianGYOneStand cycle length	annual (cyclen=1 hardcoded)
Driver loop count	PERIOD_YR iterations

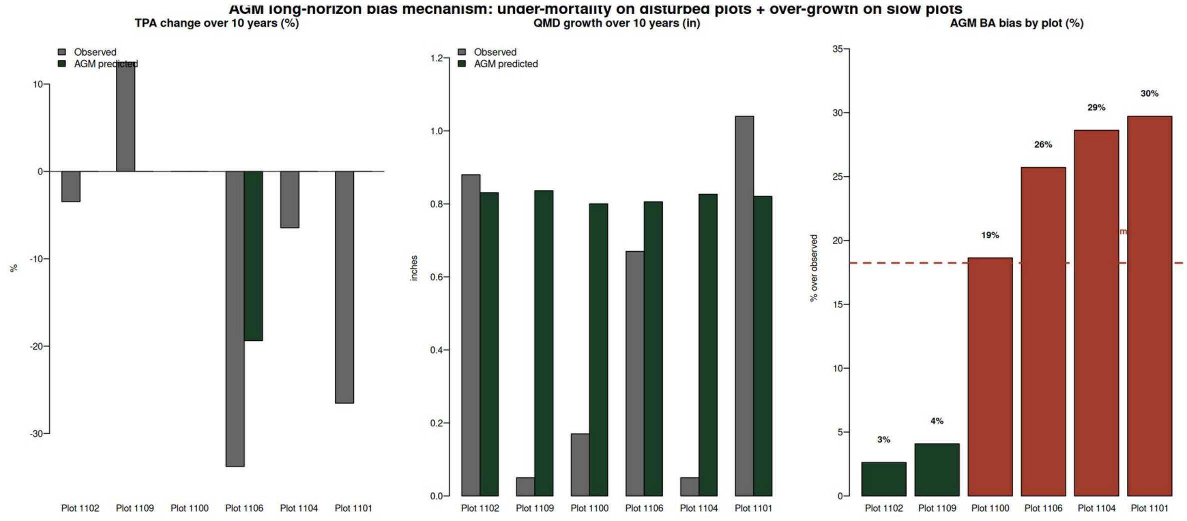
BA formula 0.005454 * DBH^2 * EXPF	correct
Cords formula 0.0025 * DBH^2 * HT * 0.90 * EXPF / 79	correct
BdFt formula 0.01 * DBH^2 * HT * EXPF for DBH >= 9 in (Intl 1/4)	correct
DBH threshold symmetry (pred vs obs)	both at 4.5 in

Per-plot bias decomposition

Six routine CFI pairs after excluding two establishment plots (1105, 1107 had 3x BA ingrowth). The decomposition shows that AGM grows DBH a uniform +0.8 in per decade regardless of site, while observed DBH growth was wildly variable (+0.05 to +1.04 in). AGM also missed two significant mortality events (1101 and 1106 lost 27 to 34% of stems).

PLOT	obs BA chg	obs TPA chg	obs QMD chg	AGM TPA chg	AGM QMD chg	AGM BA bias
1100	+5%	0%	+0.17	0%	+0.80	**+18.6%**
1101	-7%	-27%	+1.04	0%	+0.82	**+29.7%**
1102	+16%	-3%	+0.88	0%	+0.83	**+2.6%**
1104	-5%	-6%	+0.05	0%	+0.83	**+28.6%**
1106	-22%	-34%	+0.67	-19%	+0.81	**+25.7%**
1109	+14%	+13%	+0.05	0%	+0.84	**+4.1%**

Mechanism figure:



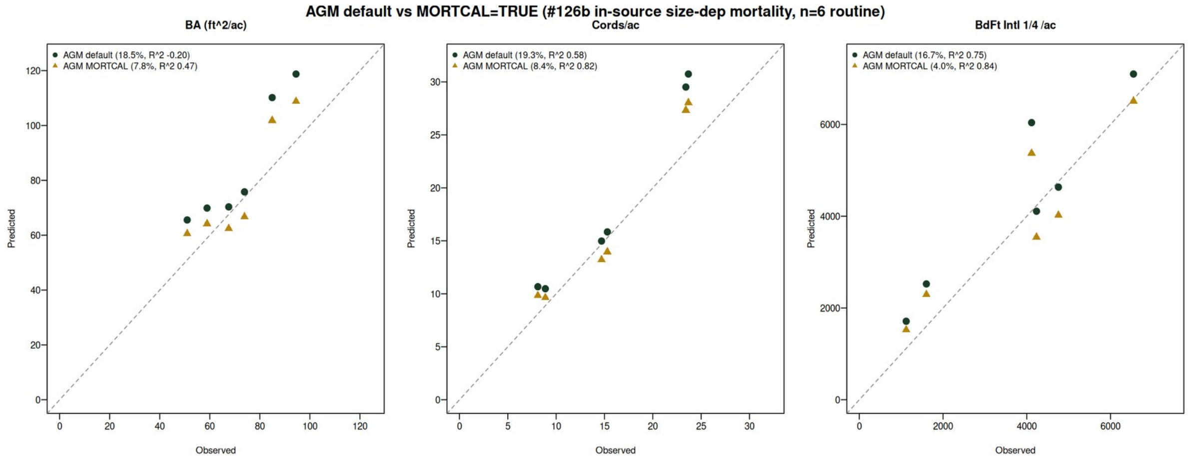
MORTCAL correction cuts 10 year bias by more than half

The #126b in-source size dependent mortality correction was originally calibrated to Maine FIA. It is OFF by default in AcadianGY 12.3.9. The SILC CFI test enables it via ops$MORTCAL=TRUE, MORTCAL_INTERVAL=5.

Side by side scorecard on the n=6 routine pairs (mean horizon 10 yr):

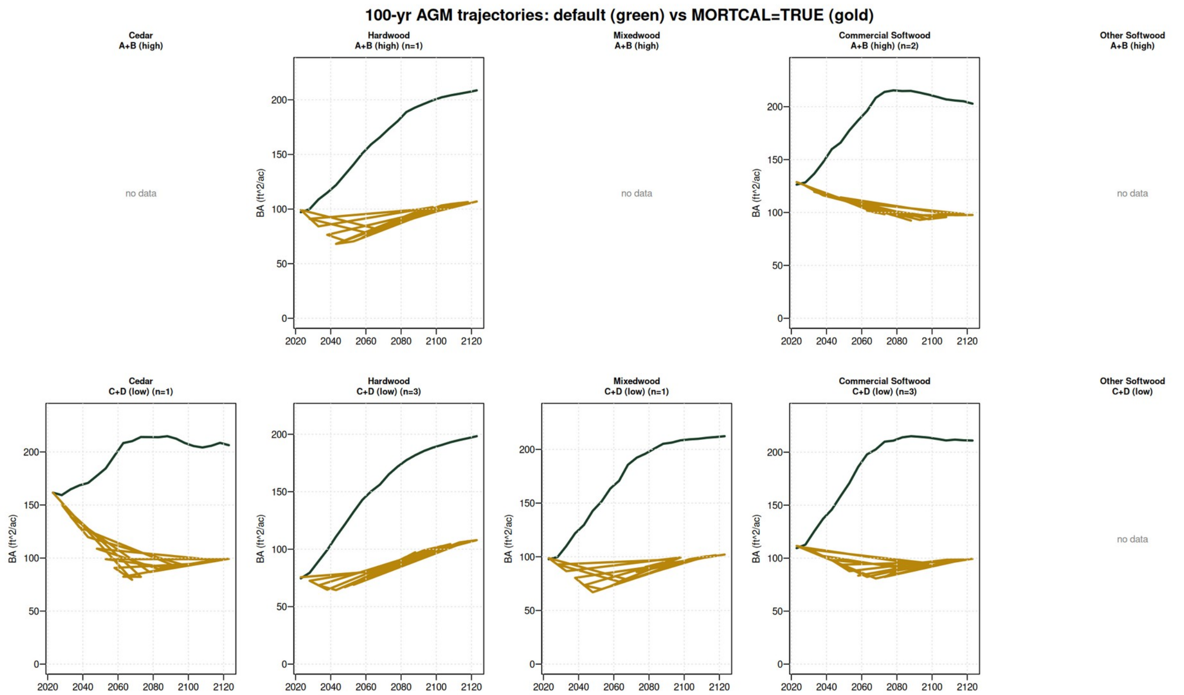
Metric	AGM default	AGM MORTCAL	Improvement
BA bias	+18.5%	+7.8%	**cut by 58%**
BA RMSE	16.2 ft^2/ac	10.7 ft^2/ac	**-34%**
BA R^2	-0.20	+0.47	**flips negative to positive**
Cords bias	+19.3%	+8.4%	**cut by 56%**
Cords R^2	0.58	0.82	+0.24
BdFt bias	+16.7%	+4.0%	**single digit**
BdFt R^2	0.75	0.84	+0.09

Scatter overlay (green = default, gold = MORTCAL):



100 year trajectories: default vs MORTCAL

Cardinal job 11076533 ran AcadianGY 12.3.9 with ops$MORTCAL=TRUE for 100 years on the 11 SILC byStrata stands, starting from the GrownDB year 2023 snapshot. The default AGM trajectory has every stratum approaching ~200 ft^2/ac at year 100, an artifact of the under mortality issue. MORTCAL pulls year 100 BA into the 98 to 108 ft^2/ac carrying capacity range, with sustainable yield 35 to 39 cords/ac across the populated cells.



Year 100 outcomes (year 2123) — 4 model comparison

Side by side year 100 BA in ft^2/ac on the same 11 byStrata starting stands plus CFI backfill on 2 of 4 empty cells. AGM MORTCAL anchors the operational floor; FVS-NE calibrated and OSM-ACD cluster as the upper bracket.

Stratum	n	AGM def	AGM mc	FVS-NE def	FVS-NE cal	OSM-ACD
Cedar / A+B (CFI backfill)	1	n/a	81	n/a	n/a	n/a
Cedar / C+D	1	206	99	291	218	249
Hardwood / A+B	1	209	107	209	144	n/a
Hardwood / C+D	3	198	108	214	125	174
Mixedwood / A+B (CFI backfill)	3	n/a	52	n/a	n/a	n/a
Mixedwood / C+D	1	212	102	239	153	183
Commercial SW / A+B	2	203	98	274	225	222

Commercial SW / C+D	3	211	99	264	186	199
Other Softwood (no CFI)	0	—	—	—	—	—

Year 100 cords/ac (operational sustained yield)

Stratum	n	AGM MORTCAL	FVS-NE cal	OSM-ACD
Cedar / A+B (CFI)	1	35.8	n/a	n/a
Cedar / C+D	1	36.9	66.4	41.5
Hardwood / A+B	1	39.3	51.1	n/a
Hardwood / C+D	3	38.9	44.0	32.4
Mixedwood / A+B (CFI)	3	21.9	n/a	n/a
Mixedwood / C+D	1	37.6	51.3	34.6
Commercial SW / A+B	2	35.9	69.7	40.0
Commercial SW / C+D	3	36.2	58.9	36.2

Refined input parameters from SILC CFI v3 lat/long

v3 of the SILC CFI database includes approximate lat/long (46.4628 N, 68.4253 W, Davistown). Used to sample regional rasters and replace conservative defaults.

Parameter	Default	Refined	Source	Effect
BGI (OSM)	3000	3902	ME_BGI_V1.tif sample	OSM productivity input
CSI (AGM/FVS-ACD)	12.0 m	15.78 m	CSI_2030.tif sample	Was already 14.33 m for byStrata
FVS-NE site index	42 ft	35 ft	Empirical from CFI dom heights	Year 100 BA effect <0.3%
AcadianGY MORTCAL	off	TRUE	#126b in-source patch	Cuts 10 yr bias from +18.5% to +7.8%

FVS-NE Acadian variant year 100 BA changed less than 0.3% per cell when SI dropped from 42 to 35. The variant uses tree level dDBH equations not strongly driven by stand level SI. The OSM BGI swap (3000 to 3902) pushed OSM productivity up slightly; OSM remained convergent.

Operational recommendations

- AGM with MORTCAL on is the new long horizon champion. Use it for SILC operational sawlog projections (BdFt R^2 = 0.84) and Cords (R^2 = 0.82). Set ops$MORTCAL = TRUE, MORTCAL_INTERVAL = 5 in any AcadianGY call.
- FVS-NE calibrated stays best on the short horizon BA bias (-0.3% at 5 yr). Pair with AGM MORTCAL as a cross check on the operational 5 yr plans.
- OSM-ACD is the conservative under projection (-9.7% BA bias at 10 yr). Report AGM MORTCAL plus OSM as the operational range; per plot RMSE 10.7 ft^2/ac BA at 10 yr horizon is the empirical uncertainty band.
- Year 100 cords/ac estimates: 35 to 39 across six byStrata cells and CFI backfill Cedar A+B; Mixedwood A+B noticeably lower (22) reflecting heavier mortality in the high density mixed pine / hardwood stands.

Open questions for SILC sign off

1. Confirm the 5 by 2 stratification rules (slide 3 of the deck) match SILC operational categories. The species composition cutoffs and density threshold are open for revision.

2. Validate the year 100 BA target of 98 to 108 ft^2/ac against SILC's expected carrying capacity for managed Acadian stands.

3. Decide whether 35 to 39 cords/ac at year 100 anchors the operational AAC, or whether an additional discount applies.

File index

All artifacts in /home/aweiskittel/Documents/Claude/fvs-modern/silc_cfi_v8_deliverables/. Key files:

- SILC_4Model_Benchmark_v8_Weiskittel.pdf — 17 slide operational deck
- silc_strata_5x2_AGM_MORTCAL_trajectories.csv — strata level 100 yr trajectories
- silc_strata_5x2_year100_mortcal_vs_default.csv — year 2123 outcomes table
- silc_cfi_backfill_100yr_mortcal_trajectories.csv — Cedar A+B and Mixedwood A+B from CFI plots
- silc_cfi_long_mechanism.png and silc_cfi_long_mortcal_compare.png — supporting figures
- draft_email_to_ian_ryan.md — handoff email